DATA ANALYTICS · 2016

Video Game Market Analysis

Insights for Strategic Campaign Planning · 2017

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CONTEXT & OBJECTIVE

The Business Question

Ice Online Store sells video games worldwide and needs to plan its 2017 advertising campaigns.

The challenge: which games, genres and platforms are most likely to succeed?

This analysis uses historical data through 2016 to answer that question.

What We Analysed

16,715 games
across multiple platforms, regions and genres

Platform lifecycle & sales trends

Regional consumer profiles (NA · EU · JP)

Critic & user review impact on commercial success

Statistical hypothesis testing

Source: Ice Online Store dataset · Python · Pandas · Matplotlib · SciPy

METHODOLOGY

A structured 5-step analytical framework

01

Data Overview

16,715 records · 11 variables · Sales across 4 regions

02

Data Preparation

Column standardisation · Type conversion · Missing value treatment · TBD handling

03

Exploratory Analysis

Platform lifecycle · Sales trends 2012–2016 · Review impact · Genre performance

04

Regional Profiles

Top platforms & genres per region · ESRB rating impact on sales

05

Hypothesis Testing

Welch t-test · $\alpha = 0.05$ · Xbox One vs PC · Action vs Sports

Tools: Python · Pandas · NumPy · Matplotlib · Seaborn · SciPy

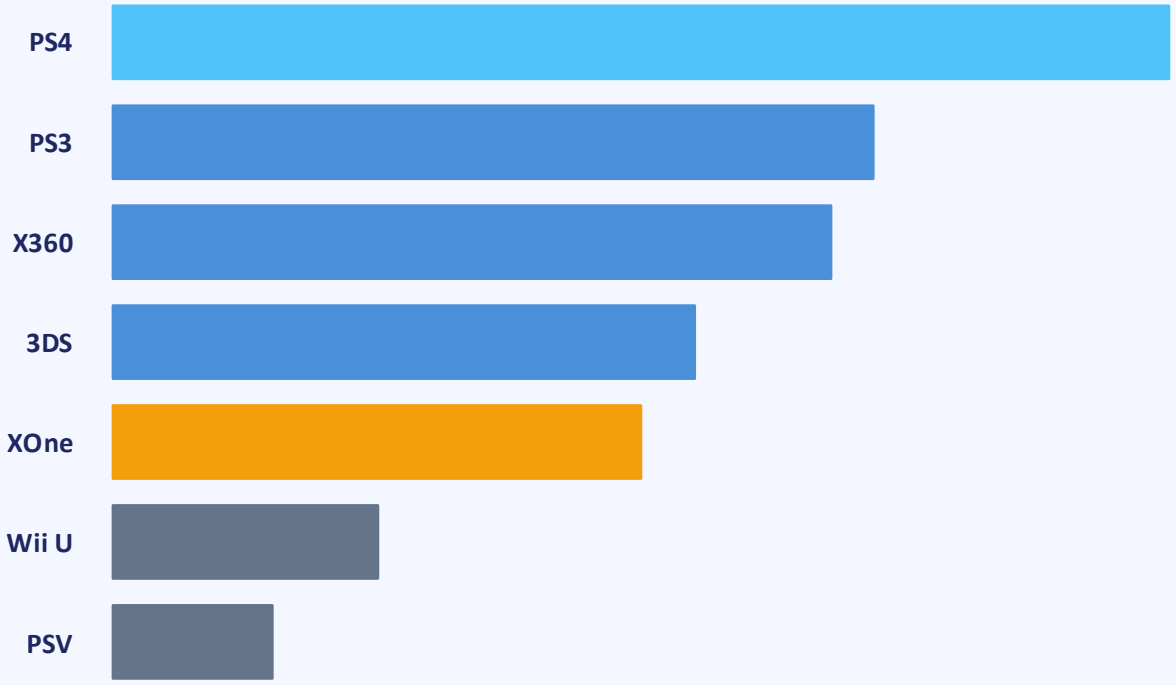
MARKET OVERVIEW · 2012–2016

16,715	2008–09	2012–16	PS4 #1
Games Analysed	Peak Launch Years	Relevant Period	Top Platform

KEY FINDING

The video game market peaked in 2008–2009 and has been consolidating since. Platforms have an average active commercial lifecycle of 5–7 years — meaning that focusing campaigns on current-generation consoles (PS4, Xbox One) is critical for 2017 success. Older platforms like PS3 and Xbox 360 are in terminal decline.

PLATFORM ANALYSIS · Sales Leaders 2012–2016



INSIGHT

PS4 leads
with clear margin, confirming its dominance in the new generation.

Xbox One
shows strong growth potential for 2017.

PS3 & X360
still relevant but declining — transitional investment only.

Index: PS4 = 100 (relative sales volume, 2012–2016)

DO REVIEWS DRIVE SALES?

Correlation analysis · PS4 platform

CRITIC SCORE

0.41

Moderate Positive Correlation

Games with higher critic scores tend to sell more — but the relationship is not deterministic. Marketing, brand recognition and distribution also play major roles.

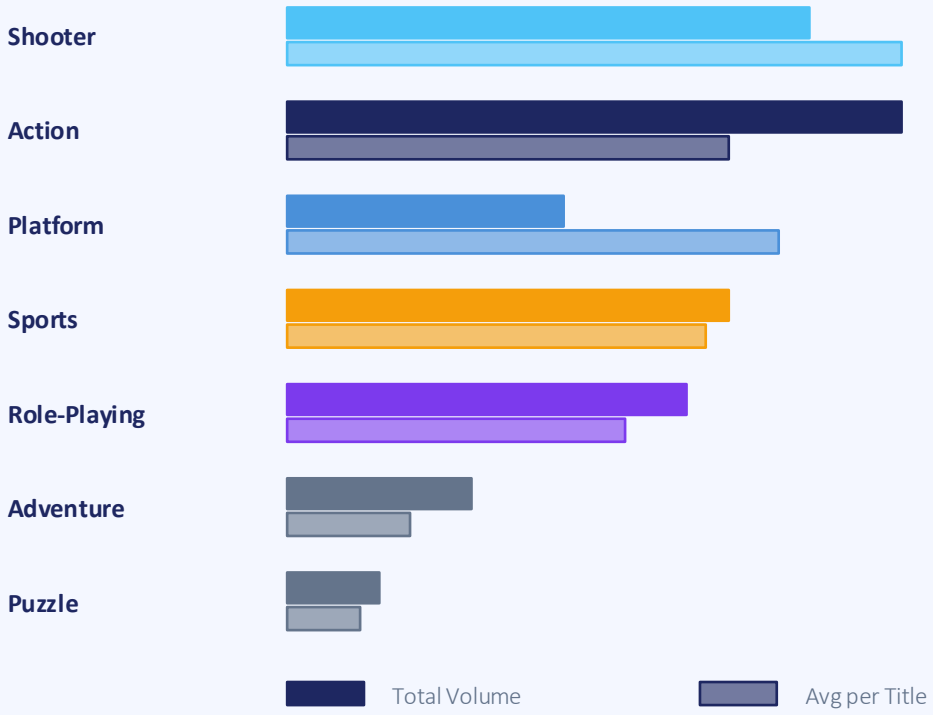
USER SCORE

-0.03

No Meaningful Correlation

User reviews show virtually no relationship with sales performance. Consumer satisfaction after purchase does not predict commercial success.

GENRE PERFORMANCE · Total vs Average Sales



STRATEGIC INSIGHT

Shooter has the highest average sales per title — the strongest commercial bet per launch.

Action dominates in total volume due to high number of releases.

Adventure & Puzzle consistently underperform — high risk for investment.

Recommendation: Prioritise Shooter and Action genres for 2017 campaign focus.

REGIONAL CONSUMER PROFILES

North America

TOP PLATFORMS

#1 Xbox 360 · PS4 · PS3

TOP GENRES

Action · Shooter · Sports

ESRB IMPACT

M (Mature) dominates

Europe

TOP PLATFORMS

#1 PS4 · PS3 · Xbox 360

TOP GENRES

Action · Shooter · Racing

ESRB IMPACT

M (Mature) dominates

Japan

TOP PLATFORMS

#1 3DS · PS3 · PSVita

TOP GENRES

Role-Playing · Action · Simulation

ESRB IMPACT

E (Everyone) leads

Key takeaway: Japan requires a distinct strategy — portable platforms and RPG genres dominate vs. console-heavy Action focus in NA & EU.

STATISTICAL HYPOTHESIS TESTING

Welch's t-test · Significance level $\alpha = 0.05$

HYPOTHESIS 1

Xbox One vs PC User Ratings

H_0: Mean ratings are equal
H_1: Mean ratings differ

p-value ≈ 0.55 → FAIL TO REJECT H_0

No statistically significant difference in user ratings between Xbox One and PC players. Platform alone does not determine user perception.

HYPOTHESIS 2

Action vs Sports User Ratings

H_0: Mean ratings are equal
H_1: Mean ratings differ

$p \approx 4.24 \times 10^{-20}$ → REJECT H_0

Highly significant difference confirmed. Action and Sports players have distinctly different expectations — requiring separate positioning strategies.

CONCLUSIONS & RECOMMENDATIONS · 2017

1

Bet on PS4 & Xbox One

Current-generation consoles concentrate the highest commercial potential. Legacy platforms (PS3, X360) warrant transitional investment only.

2

Lead with Shooter & Action

Shooter has the highest average sales per title. Action dominates total volume. Both represent the safest investment for 2017 campaigns.

3

Regional strategy is non-negotiable

Japan requires a completely different approach — portable platforms and RPG focus. NA and EU share similar profiles but differ in platform preference.

4

Invest in critical reception

Critic scores show moderate positive correlation with sales (r=0.41). Securing strong review coverage at launch can meaningfully impact performance.

Analysis period: 2012–2016 · Dataset: Ice Online Store · Tools: Python, Pandas, SciPy, Matplotlib

Thank You

Questions & Discussion

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